

The Cycle-Constrained Aerial Suppression Base and Water Point Location Problem: Joint Siting under Uncertainty with an Application to Cundinamarca, Colombia

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Abstract

Aerial suppression productivity is governed by a three-leg cycle: an aircraft positions once from base to fire, then repeats the fire-to-water refill leg on every drop, so water point location, not base-to-fire distance alone, drives delivered water. We introduce the cycle-constrained aerial suppression base and water point location problem, the first to site bases and water refill points jointly with cycle-endogenous productivity, as a two-stage mean-risk (CVaR) stochastic program with an exact linearization and a fix-and-optimize matheuristic whose reduced subproblems are exact. An open-data case study for Cundinamarca, Colombia, built around the country’s Firehawk acquisition, shows a fleet-limited-first, physics-limited-second regime: budget dominates, uncalibrated growth constants act as a containability cliff, sequential bases-first planning matches the integrated optimum wherever certification is possible, and tractability collapses exactly where fleets and budget slack grow, which the matheuristic crosses.

Keywords: wildfire suppression; facility location; aerial firefighting; water refill points; two-stage stochastic programming; conditional value-at-risk; sample average approximation; matheuristics; Colombia.

1 Introduction

Colombia’s eastern Andean department of Cundinamarca leads the country’s historical record of reported forest fires, and the extreme January 2024 fire episode around Bogotá placed aerial suppression at the center of national policy. In response, the Colombian Aerospace Force (FAC), with the national disaster risk management unit (UNGRD), committed 150,000 millones COP to acquire two Sikorsky S-70i Firehawk helicopters, the first of their class in Latin America (Fuerza Aeroespacial Colombiana, 2026). A country buying firefighting aircraft faces an immediate strategic question that current models do not answer: where should the supporting bases and, critically, the water refill points be located, and how should a single budget be split among bases, aircraft, and water infrastructure?

The operational physics of helicopter suppression make this a joint question. An aircraft flies from its base to the fire once, to position, and then cycles between the fire and the nearest usable water point for every drop. As the number of drops grows, total delivered water is

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dominated by the fire-to-water leg, which is paid on every cycle, not by the base-to-fire leg, which is paid once. Aircraft productivity is therefore endogenous to the joint configuration of bases and water points. A classical facility location model built on base-to-fire distance alone, of the p-median family (Hakimi, 1964), cannot represent this tradeoff, and, as Section 2 shows, no existing model sites bases and water points jointly with cycle-endogenous productivity.

This paper makes four contributions. First, we formulate the cycle-constrained aerial suppression base and water point location problem as a two-stage stochastic program: first-stage decisions open bases, station aircraft, and enable water points under a single budget; second-stage recourse dispatches aircraft to fires scenario by scenario. The objective is a mean-risk convex combination of expected loss and conditional value-at-risk (Rockafellar and Uryasev, 2000). Fire containment is physical: a fire is contained if the liters delivered within a critical window exceed a requirement that grows with arrival time and rate of spread, so no historical suppression outcome data, which do not exist for Colombia, are needed. The only nonlinearity, a bilinear product of dispatch and refill assignment, admits an exact McCormick linearization.

Second, we develop a fix-and-optimize matheuristic (Helber and Sahling, 2010) with large neighborhood search (Shaw, 1998) over the first-stage variables. Each iteration solves a reduced sub-MILP that is exact by construction, over a geographic neighborhood of candidate sites plus the incumbent’s open sites; a probabilistic random destroy operator, standard in adaptive large neighborhood search (Ropke and Pisinger, 2006), provably escapes a class of local optima that purely geographic neighborhoods cannot leave. The reduced sub-problem keeps the full scenario set, so every accepted move is a verified improvement for the true problem.

Third, we construct a fully sourced, replicable case study for Cundinamarca from open data only: NASA FIRMS satellite detections for all of 2024, clustered into events with ST-DBSCAN (Birant and Kut, 2007); land cover, slope, and wind from ESA WorldCover, SRTM, and NASA POWER; population exposure from WorldPop, cross-checked against the Colombian census; 120 candidate bases from the national aerodrome index and 5,449 candidate water points from OpenStreetMap and the regional water authority; and the real Firehawk’s cost, tank, and speed. Every parameter is either traced to a verified source or explicitly declared an assumption and swept in the sensitivity analysis.

Fourth, our computational study yields policy-relevant structural findings rather than only benchmark numbers. Under the real program budget the deployment is fleet-limited first and physics-limited second: one Firehawk absorbs half the budget and serves at most one fire per day-scenario, budget moves the objective most, and the uncalibrated containment constants act as a containability cliff that decides whether that aircraft buys anything at all, while site costs never move the objective across their documented ranges. A best-case sequential planner that sites bases first and water second matches the integrated optimum in every configuration we can certify, one and two aircraft alike, and the strongest integrated search available finds nothing better where certification fails; we explain structurally why. Conversely, the direct MILP loses tractability exactly where the integrated question becomes interesting, in multi-aircraft, high-budget-slack regimes, which is what motivates the matheuristic.

The remainder of the paper is organized as follows. Section 2 reviews the literature and states the gap. Section 3 formulates the model. Section 4 describes the Cundinamarca case study data. Section 5 presents the matheuristic. Section 6 reports the computational study, and Section 7 concludes with limitations and ongoing work.

2 Related work and gap

Aerial resource location for wildfire suppression began with airtanker base selection: Hodgson and Newstead (1978) chose bases with location-allocation models for one-strike initial attack, and MacLellan and Martell (1996) based airtankers for Ontario with an expected-cost model over historical demand. Modern strategic deployment models keep this base-side focus:

Sköld Gustafsson et al. (2024) locate aerial firefighting resources in Sweden with a binary program over wildfire risk data, and Mendes and Alvelos (2023) place wildland fire suppression resources on a landscape with mixed-integer programming and iterated local search. Stochastic treatments deploy suppression resources across existing stations under fire-occurrence uncertainty: the scenario-based standard-response models of Haight and Fried (2007), the simulation plus stochastic integer programming approach of Ntaimo et al. (2013), the two-stage seasonal deployment and dispatch design of Wei et al. (2015), which is the closest methodological comparator to our second stage, and the robust prepositioning of Mendes and Alvelos (2025). Recent work also allocates heterogeneous aerial fleets across capacity-constrained airbases under efficiency and equity objectives (Tasias and Nenes, 2026). In all of these, water is either absent or exogenous.

A separate stream sites water infrastructure, with the aircraft side fixed: Ramalho et al. (2021) and Ramalho et al. (2024) allocate water reservoirs by fire risk with geotechnological methods, and Alfaro et al. (2024) optimize reservoir placement for helicopter transport in Valparaíso. Operational models capture the refill cycle but take infrastructure as given: Rodríguez-Veiga et al. (2018) assign aircraft to flight routes and refueling points during suppression, Rodríguez Barreiro et al. (2024) plan helicopter circuits including water loading points for a given fire, and Skorin-Kapov et al. (2024) schedule aerial resources against zone-assigned water loading points. Closest in scope, Maraş et al. (2023) site both heliports and water sources with GIS-based multi-objective programming, but in two separate sequential deterministic models with fixed response-time thresholds, without a refill cycle and without uncertainty.

Table 1 classifies this literature against the three conditions that jointly define the gap: (a) joint siting of bases and water points in one model, (b) aircraft productivity endogenous to the base to fire to water cycle, and (c) strategic, pre-season decisions under uncertainty. Reading the table by column isolates the gap. Cycle-dependent productivity appears only at the operational level, where no siting takes place; joint siting appears in no prior model; and no work combines cycle-dependent productivity with strategic, uncertainty-aware siting. The proposed model is the only row satisfying all three, under stochastic programming with a mean-risk CVaR objective. The nearest works each own a single pillar: Maraş et al. (2023) site both facility types but sequentially, deterministically, and without a cycle; Rodríguez Barreiro et al. (2024) capture the cycle, and even select loading points, but operationally, with given infrastructure; and Wei et al. (2015) are strategic and stochastic but site no water and use coverage rather than a refill cycle.

3 Problem formulation

3.1 Setting and notation

A planner chooses, before the fire season, which candidate aerial bases to open, how many aircraft to station at each, and which candidate water refill points to enable, under a single budget. Fire days then realize randomly. On a fire day, aircraft are dispatched from open bases to fires; an aircraft serving a fire refills at exactly one enabled water point for the whole episode, and the water it delivers within the critical suppression window is determined by the three-leg cycle geometry. A fire whose delivered water reaches a physical requirement is contained; otherwise it escapes and its exposed population counts as loss.

Table 2 summarizes the notation. Distances are in meters in the national planimetric CRS (EPSG:9377), time is in hours, and money is in Colombian pesos (COP).

Table 1: Classification of the related literature against the three conditions that define the gap. Bases: the model decides the location of aerial bases or heliports. Water: the model decides the location of water refill points or reservoirs. Joint: both location decisions are made within a single integrated model. Cyclic: aircraft productivity is endogenous to the base to fire to water refill cycle within the decision model. Strategic: infrastructure or deployment decisions taken before the season or the fire day. Uncertainty: treatment of uncertainty (No, deterministic; SP, stochastic programming; RO, robust optimization). Partial indicates partial satisfaction, for example positioning resources at existing bases, or associating water demand by zone without siting refill points.

Reference	Bases	Water	Joint	Cyclic	Strategic	Uncertainty
Hodgson and Newstead (1978)	Yes	No	No	No	Yes	No
MacLellan and Martell (1996) ^a	Yes	No	No	No	Yes	No
Sköld Gustafsson et al. (2024) ^b	Yes	No	No	No	Yes	No
Haight and Fried (2007)	Partial	No	No	No	Yes	SP
Ntairo et al. (2013)	Partial	No	No	No	Yes	SP
Wei et al. (2015)	Partial	No	No	No	Yes	SP
Mendes and Alvelos (2025)	Partial	No	No	No	Yes	RO
Alfaro et al. (2024)	No	Yes	No	No	Yes	No
Ramalho et al. (2021)	No	Yes	No	No	Yes	No
Ramalho et al. (2024)	No	Yes	No	No	Yes	No
Rodríguez-Veiga et al. (2018)	No	Partial	No	Yes	No	No
Rodríguez Barreiro et al. (2024) ^c	No	Partial	No	Yes	No	No
Skorin-Kapov et al. (2024) ^d	No	Partial	No	No	No	No
Maraş et al. (2023)	Yes	Yes	No	No	Yes	No
This paper	Yes	Yes	Yes	Yes	Yes	SP

^a Minimizes expected cost over multi-season historical demand, without recourse.

^b Conference paper (ISCRAM proceedings).

^c arXiv preprint.

^d The Cyclic cell is confirmed only via a secondary paraphrase of the paywalled primary text (drop counts appear to be exogenous inputs); the remaining cells are confirmed against the openly available abstract.

3.2 Cycle productivity and physical containment

For every i, f, k, m , and s , the following quantities are parameters, computable offline, because every input to them is a parameter under the discrete candidate-site, single-aircraft-type scope:

$$\text{cyc}_{fkm} = 2\tau_{fk}^{\text{fw}} + \delta, \quad (1)$$

$$d_{ifkm} = \max\left(0, \left\lfloor \frac{W - \tau_{if}^{\text{bf}}}{\text{cyc}_{fkm}} \right\rfloor\right), \quad (2)$$

$$L_{ifkm} = d_{ifkm} Q_m, \quad (3)$$

so L_{ifkm} is the liters one aircraft of type m delivers to fire f when dispatched from base i and refilling at point k : it flies the positioning leg once, then completes as many refill cycles as fit in the remaining window. A base too far to complete even one round trip contributes zero drops. This is the formal core of the cycle insight: τ_{if}^{bf} enters once, through the numerator, while τ_{fk}^{fw} divides the window, so water geometry scales productivity multiplicatively. Three conventions are implicit in the drop count and disclosed here: the aircraft is assumed to arrive at the fire with an empty tank, so its first drop, like every later one, costs a full refill cycle (an aircraft departing loaded would gain at most one drop; we take the conservative count); a single cruise speed covers loaded and unloaded legs, matching the single sourced speed specification; and flight time carries no operating cost in the objective, so dispatch is limited by the fleet and the budget, never by fuel accounting.

Table 2: Notation. Every fire-indexed quantity is scoped to its scenario: a bare f means for $f \in F_s$ within scenario s .

<i>Sets</i>	
I	candidate aerial base sites, index i
K	candidate water refill points, index k
M	aircraft types, index m (a single type in this study)
S	SAA scenario sample, index s , probability p_s
F_s	fires in scenario s , index f
<i>Parameters</i>	
B	total first-stage budget, COP
$c_i^{\text{base}}, c_m^{\text{air}}, c_k^{\text{wat}}$	cost to open base i , acquire one aircraft of type m , enable point k
Q_m	tank capacity of type m , liters per drop
v_m	flight speed of type m , meters per hour
W	critical suppression window length, hours
δ	per-cycle operational overhead (fill plus drop), hours
$\tau_{if}^{\text{bf}}, \tau_{fk}^{\text{fw}}$	one-way base-to-fire and fire-to-water travel times, hours
D_f	value at risk of fire f (population exposure)
ρ_f	rate-of-spread parameter of fire f
t_f^{arr}	earliest-attack arrival time of fire f , hours (Section 3.2)
A_0, c	initial fire area and liters per square meter to control
α, λ	CVaR confidence level and mean-risk weight
<i>First-stage variables</i>	
$y_i \in \{0, 1\}$	1 if base i is opened
$w_k \in \{0, 1\}$	1 if water point k is enabled
$n_{im} \in \mathbb{Z}_+$	aircraft of type m stationed at base i
<i>Second-stage variables (per scenario s)</i>	
$x_{ifms} \in \mathbb{Z}_+$	aircraft of type m sent from base i to fire f
$z_{fks} \in \{0, 1\}$	1 if fire f refills at water point k
$q_{fs} \geq 0$	liters delivered to fire f
$e_{fs} \in \{0, 1\}$	1 if fire f is not contained
$\ell_s \geq 0$	scenario loss
$\eta \in \mathbb{R}, t_s \geq 0$	CVaR auxiliaries
$u_{ifkms} \geq 0$	linearization auxiliary (Section 3.3)

Containment is physical. Fire f is contained if delivered liters reach

$$R_f = c A_0 e^{\rho_f t_f^{\text{arr}}}, \quad (4)$$

an exponential area growth form adopted as a phenomenological simplification for tractability, with c in liters per square meter and A_0 in square meters (A_0 is quoted in hectares throughout the paper and converted exactly once in the implementation). We disclose its provenance precisely: exponential fire area growth is documented in structural fire science (Ramachandran, 1986); wildland-specific growth models are typically elliptical (Van Wagner, 1969) and empirical growth varies by fuel and weather, so Equation (4) is a modeling choice that fits a single scalar ρ_f per fire, not a wildland-validated law. Its constants A_0 and c have no Colombia-specific calibration source and are treated as swept sensitivity parameters throughout (Section 6.6). The arrival time t_f^{arr} is defined exogenously as the travel time from the closest candidate base, $\min_{i \in I} \tau_{if}^{\text{bf}}$, so that R_f remains a pure per-fire parameter independent of the dispatch decision.

The requirement and the drop count deliberately run on two different clocks, which we state explicitly. R_f sizes the fire at the earliest feasible attack time, detection plus $\min_i \tau_{if}^{\text{bf}}$; the drop count d_{ifkm} measures the window from dispatch, charging each base its full positioning leg. When the dispatching base is not the closest candidate, the fire has grown for longer than t_f^{arr} , so the requirement that dispatch faces is understated; we accept this conservative-window,

optimistic-requirement pairing rather than endogenize it, because a dispatch-dependent requirement would become R_{if} and couple the containment threshold to the recourse decision, breaking the parameter-only precomputation that keeps the model linear.

3.3 Two-stage mean-risk formulation

With $M_m = \lfloor B/c_m^{\text{air}} \rfloor$ a budget-derived fleet bound, the model is:

$$\min (1 - \lambda) \sum_{s \in S} p_s \ell_s + \lambda \left(\eta + \frac{1}{1 - \alpha} \sum_{s \in S} p_s t_s \right) \quad (5)$$

$$\text{s.t. } \sum_{i \in I} c_i^{\text{base}} y_i + \sum_{i \in I} \sum_{m \in M} c_m^{\text{air}} n_{im} + \sum_{k \in K} c_k^{\text{wat}} w_k \leq B, \quad (6)$$

$$n_{im} \leq M_m y_i, \quad \forall i, m, \quad (7)$$

$$z_{fks} \leq w_k, \quad \forall f, k, s, \quad (8)$$

$$\sum_{k \in K} z_{fks} \leq 1, \quad \forall f, s, \quad (9)$$

$$\sum_{f \in F_s} x_{ifms} \leq n_{im}, \quad \forall i, m, s, \quad (10)$$

$$q_{fs} = \sum_{i \in I} \sum_{k \in K} \sum_{m \in M} L_{ifkm} u_{ifkms}, \quad \forall f, s, \quad (11)$$

$$u_{ifkms} \leq x_{ifms}, \quad \forall i, f, k, m, s, \quad (12)$$

$$u_{ifkms} \leq M_m z_{fks}, \quad \forall i, f, k, m, s, \quad (13)$$

$$u_{ifkms} \geq x_{ifms} - M_m(1 - z_{fks}), \quad \forall i, f, k, m, s, \quad (14)$$

$$q_{fs} \geq R_f(1 - e_{fs}), \quad \forall f, s, \quad (15)$$

$$\ell_s = \sum_{f \in F_s} D_f e_{fs}, \quad \forall s, \quad (16)$$

$$t_s \geq \ell_s - \eta, \quad \forall s, \quad (17)$$

$$y_i, w_k, z_{fks}, e_{fs} \in \{0, 1\}, \quad n_{im}, x_{ifms} \in \mathbb{Z}_+, \quad q_{fs}, \ell_s, t_s, u_{ifkms} \geq 0, \quad \eta \text{ free.}$$

The objective (5) is the mean-risk convex combination of expected loss and CVaR in the linear reformulation of Rockafellar and Uryasev (2000); $\lambda = 0$ is risk neutral and $\lambda = 1$ is pure CVaR. Constraint (6) is the single budget covering bases, fleet, and water points, which makes the budget split endogenous. Constraints (7) to (10) link stationing to open bases, refills to enabled points, one refill point per served fire, and dispatch to the stationed fleet. The delivered liters coupling is bilinear in the natural formulation, $q_{fs} = \sum L_{ifkm} x_{ifms} z_{fks}$; constraints (12) to (14) replace each product with the auxiliary u_{ifkms} through a McCormick envelope that is exact, not a relaxation, because z_{fks} is binary and $x_{ifms} \leq M_m$ is valid. Constraint (15) enforces physical containment, and (16) to (17) define scenario loss and the CVaR linking. Scenario probabilities must sum to one; the mean-risk CVaR objective is otherwise unbounded, a subtle failure mode we document because it silently arises when subsampling scenario sets without renormalizing.

Section 6.1 verifies computationally that the linearized model and the literal bilinear program solved with a nonconvex MIQCP solver return identical optimal values on real instances. The auxiliary count $|I| \times |K|$ per fire and scenario is what makes the full 5,449-point water catalog intractable for the direct MILP and motivates the matheuristic of Section 5.

The formulation treats each scenario as a single short episode with no aircraft reuse across fires over time: constraint (10) caps total dispatches per scenario at the stationed fleet. With day-length scenarios and a two-hour window this understates what one aircraft could fly in a full day, a deliberate simplification whose consequences we quantify and discuss in Sections 6.6 and 7.

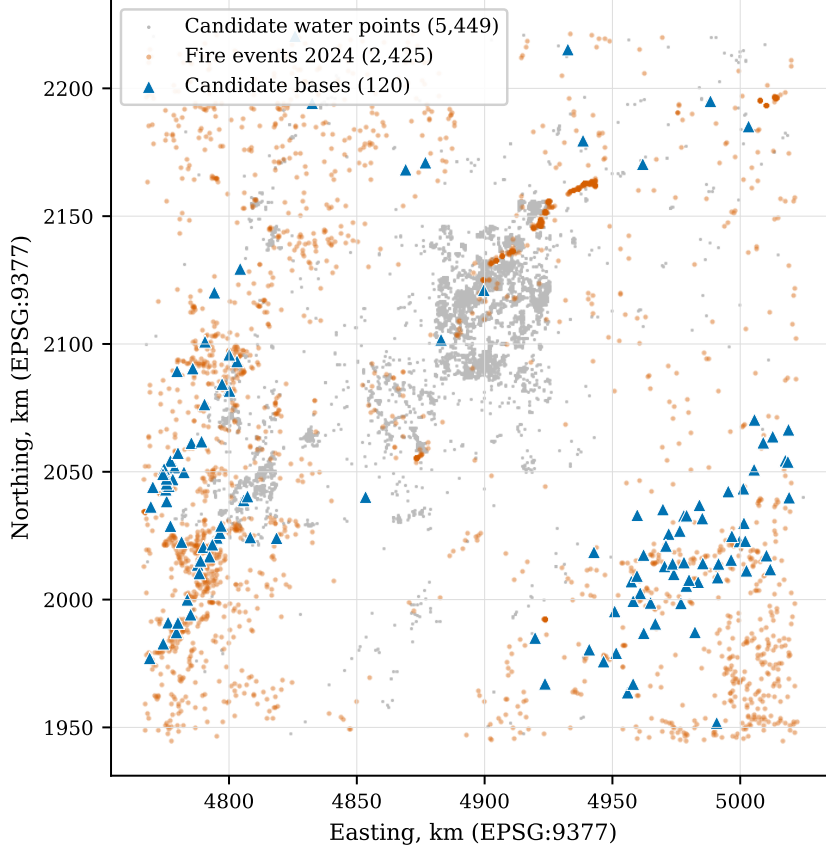


Figure 1: The Cundinamarca instance: 2,425 fire events detected during calendar year 2024, 120 candidate aerial bases, and 5,449 candidate water refill points, in the national planimetric CRS (EPSG:9377).

4 Case study: Cundinamarca, Colombia

All data are open, and every processing decision is scripted in the companion repository. Figure 1 shows the study area.

4.1 Fire events and scenarios

Fire events come from NASA FIRMS VIIRS 375 m active fire detections (NASA FIRMS, 2026) for the full calendar year 2024 over a Cundinamarca bounding box: 4,719 raw detections reduce to 4,470 after coordinate validation, exact deduplication, and a nominal-confidence filter, and are clustered into 2,425 events with ST-DBSCAN (Birant and Kut, 2007), using $\varepsilon_{\text{spatial}} = 850$ m, anchored on the 375 m sensor resolution, and $\varepsilon_{\text{temporal}} = 1.5$ days, anchored on revisit and persistence; sensitivity of the event catalog to both radii is part of the pipeline. Detections are projected to EPSG:9377, the national single-zone planimetric standard, before clustering. Our implementation adapts Birant and Kut (2007) in the way standard across the ST-DBSCAN software ecosystem: the original non-spatial attribute threshold becomes a continuous temporal radius intersected with the spatial one, and border points join the first cluster that reaches them.

A scenario is one historical calendar day, matching the stochastic unit of Wei et al. (2015): the 366-day pool of 2024 (282 days with at least one event) is bootstrap-resampled with replacement into $|S| = 200$ equiprobable day scenarios under sample average approximation (Kleywegt et al., 2002). Day scenarios keep the episode structure of constraint (10) valid and

assume fleet availability resets daily; cross-day dynamics are not modeled. The satellite record misses cloud-obscured fires and detects active burning rather than ignition, caveats inherited by any FIRMS-based catalog.

4.2 Fire attributes

Each fire’s rate-of-spread parameter follows a land cover base rate modified by slope and wind,

$$\rho_f = \rho_{\text{scale}} \cdot r_{\text{class}(f)} \cdot e^{0.05 \text{slope}_f} \cdot (1 + 0.1 \text{wind}_f), \quad (18)$$

with land cover class from ESA WorldCover 10 m (ESA, 2021), slope in degrees from SRTM 30 m (NASA, 2013) by Horn’s method (Horn, 1981), and wind in meters per second from NASA POWER (NASA POWER, 2026) at each fire’s location and date. The relative rates r_{class} are ordinally grounded in the fire behavior fuel models of Anderson (1982), grass fastest, shrub intermediate, timber slower, non-burnable near zero; their absolute magnitude has no Colombian calibration source, so the overall scale ρ_{scale} is a swept sensitivity parameter, and the exponential slope response follows Rothermel (1972) in form with an illustrative coefficient. The value at risk D_f is the WorldPop 2020 constrained population count (WorldPop, 2020) within 1 km of the fire, an unmonetized exposure index cross-checked at municipal level against the 2018 census (DANE, 2018).

4.3 Candidate sites and aircraft

The 120 candidate bases combine the national civil aerodrome index (Aerocivil, 2025) with military installations; the 5,449 candidate water points combine named OpenStreetMap water bodies (OpenStreetMap contributors, 2026) with the regional water authority’s lagoon layer (CAR Cundinamarca, 2026). The aircraft is the Sikorsky S-70i Firehawk that Colombia is acquiring (Fuerza Aeroespacial Colombiana, 2026): 75,000 millones COP per unit (the program price of 150,000 millones for two), a 1,000 gallon (3,785 liter) tank, snorkel refill in under one minute from water at least 45 cm deep, which is what makes lagoon-scale water points usable, and a 150 knot cruise (277.8 km per hour). The per-cycle overhead δ is the sourced sub-minute refill plus an assumed 30 second drop and positioning allowance.

4.4 Parameter decisions and disclosed assumptions

Table 3 lists the base-case values of the governing parameters and the provenance of each. Three are evidence-anchored single values; site costs and the fire growth constants have no Colombia-specific source and are declared illustrative sweep ranges, never calibrated values, a disclosure discipline we consider a methodological contribution of the case study.

5 Solution method

The direct MILP is tractable only for truncated candidate sets: the linearization adds $|I| \times |K|$ auxiliaries per fire and scenario, over 650,000 per fire-scenario pair at the full catalog. We therefore solve the full problem with a fix-and-optimize matheuristic (Helber and Sahling, 2010), a MIP-based improvement scheme distinct from local branching (Fischetti and Lodi, 2003), with large neighborhood search (Shaw, 1998) over the first-stage variables.

The incumbent is a first-stage solution (y, w, n) . Each iteration selects a neighborhood of candidate sites to free, builds a reduced model containing only the freed sites plus the incumbent’s open sites, fixes the non-freed open sites at their incumbent values, solves the reduced sub-MILP exactly over the full scenario set, and accepts the result if it strictly improves the incumbent. Dropping a site that is closed in the incumbent and outside the neighborhood

Table 3: Base-case parameters, provenance, and sweep treatment. Citations for each source appear in the accompanying text.

Parameter	Base case	Provenance	Sensitivity
Budget B	150,000 M COP	real Firehawk program spend	swept 80 to 300 G
Window W	2.0 h	NWCG initial attack standard	swept 1, 2, 4, 8 h
CVaR level α	0.95	standard in the CVaR literature	exp. 4 varies risk
Risk weight λ	0.5	midpoint convention	swept 0 to 1
Aircraft cost, tank, speed	Firehawk values	sourced (FAC program)	fixed
A_0	1 ha	NWCG size-class anchor ^a	swept 0.1 to 14 ha
c	3 L/m ²	no verified source	swept 1.5 to 10
ρ_{scale}	1.0	no source	swept 0.5 to 2
c_i^{base} (uniform)	1,000 M COP	US benchmarks, scaled	swept 300 to 6,000 M
c_k^{wat} (uniform)	50 M COP	US benchmarks, scaled	swept 15 to 300 M

M denotes millones (millions); G denotes thousands of millones (10^9 COP). Agency initial-attack standards genuinely differ, from two-hour containment (National Wildfire Coordinating Group, 2026) to next-morning control (Korkola et al., 2024), hence the window sweep.

^a The A_0 range runs from the NWCG size class A bound (0.1 ha), through a size class B fire at the base case (1 ha), up to the area of a single VIIRS 375 m pixel (14 ha), the sensor’s localization bound rather than a typical fire size; none of these is a Colombian calibration, which is why the axis is swept.

is exact, not an approximation: such a site is mathematically identical to one with its binary fixed at zero. Every accepted iteration is therefore a verified improvement of the true objective, and the sub-MILP size is bounded by the neighborhood, not the catalog.

Neighborhood selection follows Shaw’s relatedness principle in geographic form: a random seed site anchors each iteration, and the nearest candidate bases and water points to it are freed together, matching the problem’s own coupling, since sites an aircraft would cycle between are exactly the sites worth reconsidering jointly. Pure geographic neighborhoods, however, provably admit local optima that no number of iterations escapes: when the only improving move must close an open base and open a distant one within the same iteration, for example when the budget affords a single aircraft so relocation is the only option, no single seed’s nearest-sites neighborhood ever contains both. We observed exactly this on a real instance, where the optimal base lay 174 km from the incumbent’s and 200 geographic iterations produced no improvement. Following standard adaptive large neighborhood search practice (Ropke and Pisinger, 2006), each iteration is therefore, with probability π , a random destroy that frees sites drawn uniformly from the whole catalog; unit tests reproduce the trap instance and confirm the random operator escapes it reliably across seeds. A parameter sweep on the base-case instance with a known optimum (24 configurations, five seeds each, 100 iterations, early stopping disabled so reliability is not confounded with patience) discriminates sharply: pure geographic destroy missed the optimum in three of its six configurations (hit rates 0.6 to 0.8, one run stalling far above the optimum), while configurations with destroy probability 0.2 or higher reached it in 89 of 90 runs. Reliability, not only the synthetic trap, mandates the random operator. Since every mixed configuration attains the optimum, neighborhood sizes are chosen for runtime, five bases and ten water points per iteration, the fastest reliable configuration at about 22 seconds per 100-iteration run, and $\pi = 0.3$ sits inside the reliable band.

6 Computational study

All experiments run on an Intel Core i7-13620H with 16 GB RAM under Windows 11, with Gurobi 13.0.2 as the MIP solver (PuLP 3.3.2 as the modeling layer). Unless stated otherwise, instances use the base-case parameters of Table 3, candidate subsets are taken in catalog order, and scenario subsets renormalize probabilities. Two reading conventions apply throughout. Arrival times t_f^{arr} are recomputed over each instance’s truncated candidate base subset, so the

Table 4: Experiment 1: linearized MILP versus literal bilinear formulation, two real scenarios (15 fires), base-case parameters. Objectives are identical to solver precision at every size.

$ I $	$ K $	MILP obj.	MILP s	Bilinear obj.	Bilinear s	Match
3	5	8576.834	0.02	8576.834	0.02	yes
5	10	8576.834	0.11	8576.834	0.10	yes
8	15	8576.834	0.39	8576.834	0.29	yes

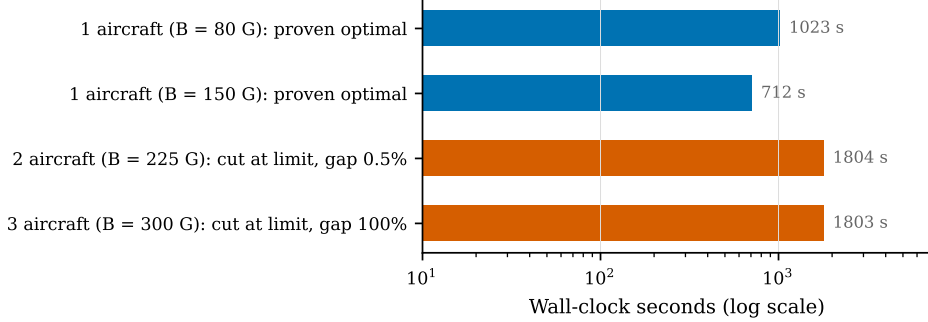


Figure 2: Direct integrated MILP wall-clock time at 20 bases, 50 water points, 81 fires, as the fleet regime grows. Blue: proven optimal. Vermillion: cut at the 1,800 s limit, labeled with the remaining MIP gap.

requirement is defined consistently within every instance. And because c_k^{wat} enters only the budget constraint, a solution with slack budget carries alternate-optimal open water points at no objective cost, so the count of open water points in any reported solution is bookkeeping noise, never a finding. Every table and figure is generated by a committed script from a committed results file.

6.1 Exactness of the linearization

Table 4 compares the linearized MILP against the literal bilinear formulation, solved by Gurobi as a nonconvex MIQCP, on real instances of increasing size. Optimal objectives match exactly in every case, confirming computationally that the McCormick envelope is exact. At these sizes the bilinear arm is somewhat faster, so we make no speed claim for the linearization; the presolve of a modern solver evidently linearizes the binary product internally at least as well. The linearized model remains the formulation of record because it is solver-portable and is the sub-problem the matheuristic solves.

6.2 Tractability and the matheuristic

Figure 2 shows how direct tractability degrades as the fleet regime grows. Single-aircraft instances at 20 bases, 50 water points, and 10 scenarios (81 fires) solve to proven optimality in 12 to 17 minutes. The two-aircraft regime is borderline: at a tight budget (160 G, leaving 8 G of slack) the direct solve proves optimality in about ten minutes, while with more slack (225 G) it still carries a 0.5 percent gap at the 1,800 s limit. At three aircraft (300 G) that limit leaves an essentially 100 percent gap, a lower bound near zero; an earlier unlimited two-aircraft probe, run before the containment units were corrected, had already burned 16 wall-clock hours without proof, which is what fixed the 1,800 s protocol used throughout. What drives the cliff is budget slack and the configuration symmetry it buys as much as fleet size itself, cheap bases alone leave a 69 percent gap at the same limit (Section 6.6), and this is precisely the regime where the integrated question matters.

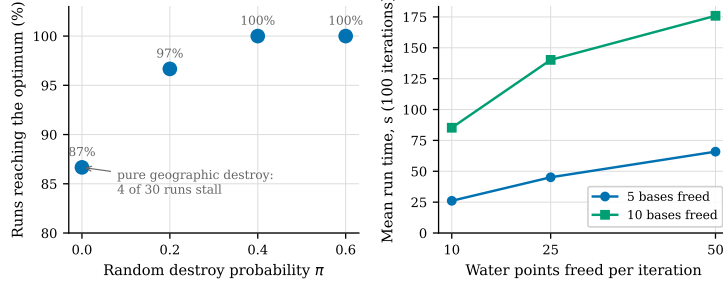


Figure 3: Matheuristic tuning sweep (24 configurations, five seeds each, 100 iterations, early stopping disabled; 20 base, 50 water point instance with a known optimum). Left: reliability. Share of the 30 runs per destroy probability π that reach the optimum; pure geographic destroy ($\pi = 0$) stalls in 4 of 30 runs. Right: cost. Mean run time by neighborhood size, averaged over destroy probabilities; cost grows in both neighborhood dimensions.

The matheuristic is the exact-by-construction alternative at catalog scale, where the direct MILP cannot even be built. On truncated instances it recovers the proven optimum at every size, provided its iteration budget is adequate: with 150 iterations (patience 60) it matches the certified optimum exactly at all three sizes, in 34 seconds at 20 bases and 50 water points, whereas a short budget (30 iterations, patience 15) stalls 10 percent above it, evidence that the corrected instance genuinely discriminates solution quality. On the full 120-base, 5,449-water-point catalog its incumbents are upper bounds without optimality certificates, and they tell the cycle story directly. With two scenarios the full catalog reaches 73.85 in 18 seconds, roughly half the proven optimum of the largest certifiable truncation (152.37 at 20 bases and 50 water points): the denser water network buys containment that no truncation can. With ten scenarios (81 fires) the full-catalog incumbent lands within solver tolerance of the certified truncated optimum (22,243 against 22,242, at a different, alternate-optimal base) in 235 seconds. Figure 3 summarizes the tuning sweep behind Section 5’s configuration choices: the random destroy operator is what makes the search reliable, and sub-solve cost scales multiplicatively in both neighborhood dimensions, consistent with the $|I| \times |K|$ auxiliary count.

6.3 Integrated versus sequential siting

The star comparison asks what joint optimization is worth against a sequential planner who sites bases first and water second. Our sequential baseline is deliberately strong. Phase A solves the same model with a water-blind relaxation, every fire is assumed to reach its best-case water point in the full catalog at zero cost, under a budget share ϕB for bases and fleet; phase B fixes phase A’s bases and fleet and solves the full model, real water catalog and costs, real total budget, deciding only water points and recourse. We sweep the split ϕ and report the best sequential result, so the comparison is against a best-case sequential planner, not a straw man. By construction the sequential solution is feasible for the integrated model, so it can never win; on a hand-built instance in the companion test suite it is strictly worse at every ϕ , because a water-blind phase A commits its single affordable aircraft toward a marginally more valuable fire whose only water point is economically out of reach, a misdirection no budget split can undo.

On the real Cundinamarca instances the result is an honest null with a sharp structural explanation. Across every configuration tested, single-aircraft budgets of 78, 80, 90, and 150 G COP, water costs at both 50 M and the top-of-range 300 M, two and ten scenarios, up to 20 bases and 50 water points, and now also the two-aircraft budget of 160 G, which the corrected instance solves to proven optimality in about ten minutes, the best sequential solution matches the integrated optimum to solver tolerance (at 90 G the sequential value even lands 0.005

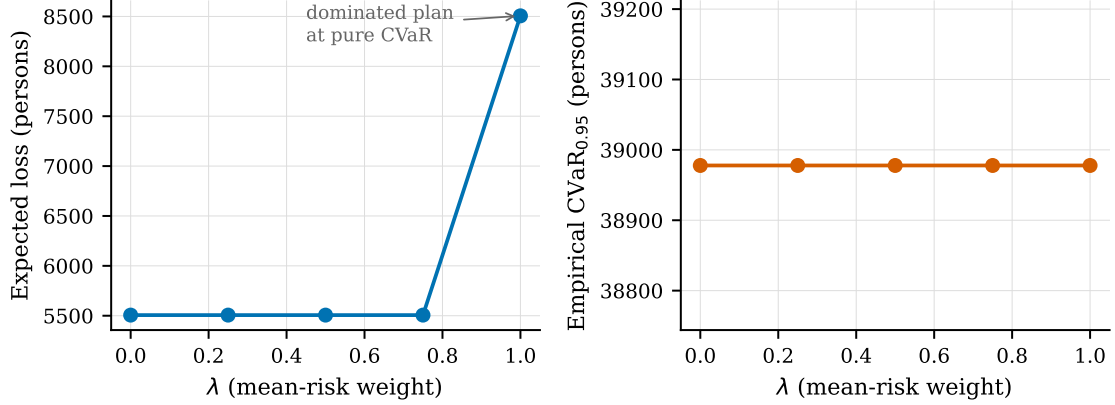


Figure 4: Experiment 4: risk-attitude sweep. Left: expected loss. Right: empirical CVaR_{0.95}, recomputed identically for every λ from each solution’s loss distribution. The frontier is degenerate under a single affordable aircraft, and pure CVaR ($\lambda = 1$) yields a dominated plan.

percent below the integrated incumbent, which stopped at the solver’s default relative gap; tolerance noise, not a real gap). The reason is regime structure: with $c^{\text{air}} = 75$ G, any budget in $[76, 152]$ G fields exactly one aircraft; constraint (10) then serves at most one fire per scenario, so at most $|S|$ distinct water points are ever needed, and even the tightest leftover, 2 G at $B = 78$ G, buys the handful that matter. Water cannot compete for budget, and a water-blind base choice is as good as an informed one. The ϕ sweep exposes the mechanism directly: every split that cannot fund the aircraft ($\phi < 0.975$ at 78 G) collapses to the all-escape plan, and every split that can reproduces the same optimum. The baseline also recommends itself as a warm start, reaching the 160 G optimum in 36 seconds against 618 for the direct integrated solve. The honest headline is conditional: sequential siting is optimal or indistinguishable from optimal in every regime we can certify, one and two aircraft alike; the hand-built counterexample shows the gap becomes strictly positive when a single decisive water point is economically out of a water-blind planner’s reach; and the regime where the integrated model plausibly matters, three-plus aircraft under large budget slack, is exactly the one the direct MILP cannot certify (Sections 6.2 and 6.6). We therefore measured that regime against the strongest integrated search available instead of leaving it bounded: at 300 G the matheuristic, run cold and re-run warm-started from the sequential plan itself (150 iterations, about half an hour each), lands on the sequential value to solver tolerance, as does the direct solver’s own 1,800 s incumbent, four independent solution paths agreeing at 14,478.11. What is missing in the three-aircraft regime is the certificate, whose lower bound configuration symmetry destroys, not a better plan.

6.4 Risk attitude

Figure 4 sweeps the mean-risk weight λ on the 20-base, 50-water, 10-scenario instance. The efficient frontier is degenerate: expected loss and CVaR are both flat from $\lambda = 0$ to 0.75, because a single affordable aircraft cannot hedge the worst day at all, there is no tail-protection to buy. At $\lambda = 1$, pure CVaR, the solver returns a dominated plan: identical CVaR, 54 percent worse expected loss, six more escaped fires, since a pure tail objective is indifferent to everything off the tail; every plan with the same tail value is optimal at $\lambda = 1$, and which one the solver returns is arbitrary, exactly the degeneracy a lexicographic or epsilon-regularized objective would remove. This is the textbook argument for the mean-risk combination over pure CVaR, reproduced on real data. With ten equiprobable scenarios, CVaR_{0.95} equals the worst-scenario loss, the expected small-sample behavior under SAA, which the replication analysis of Section 6.5 quantifies.

6.5 SAA replication and optimality gap

How much of the ten-scenario results is sampling noise? We follow the replication protocol of Kleywegt et al. (2002): $M = 10$ independent bootstrap replications of $N = 10$ day-scenarios each, all ten solved to proven optimality within the limit and none excluded, whose mean optimal value estimates a statistical lower bound on the true optimum, 10,505 with a 95 percent t -interval of $\pm 6,074$. The interval is wide because day-to-day fire load varies enormously (replication optima range from 41 to 22,716), which is exactly the variance a day-scenario model must carry. The ten candidate first-stage plans, one per replication, are then evaluated on an independent 100-scenario sample; the best is re-evaluated on a second, fresh 100-scenario sample, so that selecting the minimum does not flatter the bound. The selected plan's fresh-sample mean-risk value is 10,876 (expectation component $2,631 \pm 1,008$; CVaR component 19,121, a point estimate), giving an optimality-gap point estimate of 371, or 3.4 percent of the upper bound.

Two further readings matter. First, selection is what makes the bound tight: the pre-registered naive candidate, the first replication's plan, which happened to be optimized against an unusually quiet draw, evaluates to a 69 percent gap, while four of the ten independently drawn replications evaluate to the identical mean-risk value on the selection sample, so SAA at $N = 10$ already stabilizes the first-stage deployment even though it cannot certify it. Second, the CVaR component is noisy at this evaluation size: the same selected plan's sample CVaR moved from 35,947 on the selection sample to 19,121 on the fresh one, which is why the reported upper bound comes from the fresh sample only and why the gap carries a formal confidence interval only on its expectation component. The downward small-sample bias of sample CVaR loosens, without invalidating, the statistical lower bound.

6.6 Sensitivity: fleet first, containment physics second

A one-at-a-time sweep around the base case varies seven axes: budget, window, A_0 , c , ρ_{scale} , and both site costs, 21 solves in total, each under a 1,800 s limit; four points did not reach proof and are reported as incumbents with their remaining gaps (the two multi-aircraft budgets, the four-hour window, and the cheapest base cost). The result has a clear hierarchy.

Budget moves the objective most (Figure 5): 225 G fields two aircraft and improves the base case by 18 percent (incumbent at a 0.5 percent gap), and 300 G fields three for a 35 percent improvement, though there the remaining gap is essentially 100 percent, the tractability cliff of Section 6.2.

The containment constants now matter too, and they matter as a cliff rather than a dial. The base case contains 7 of 81 fires: one aircraft, one fire per day-scenario, and not every day's best fire is containable. Halving the window to one hour loses containments (objective up 6 percent); stretching it to four hours gains them (incumbent 16 percent below the base case, at a 2.5 percent gap) and then saturates, since the eight-hour point, solved to proven optimality, lands on exactly the same value: once the served fire's requirement fits inside the window, constraint (10), not time, is what binds. A_0 and c sweep the system across its containability cliff: at $A_0 = 5$ ha, or 14 ha, or $c = 10$ L/m², essentially nothing is containable by a single Firehawk (80 to 81 of 81 fires escape and the objective sits at or within 0.05 percent of the all-escape value, with the 14 ha solution opening nothing at all), while at $A_0 = 0.1$ ha or $c = 1.5$ the objective saturates at the same floor as the long windows, the value of serving the best containable fire each day. Doubling ρ_{scale} raises expected loss by 20 percent while leaving the worst day untouched: spread rate degrades the average, not the tail this fleet cannot hedge anyway.

Site costs remain the one axis that never moves the objective across its full documented range, the surviving piece of the pre-correction reading. Cheap bases do, however, destroy tractability rather than economics: at 300 M per base the solver still carried a 69 percent gap

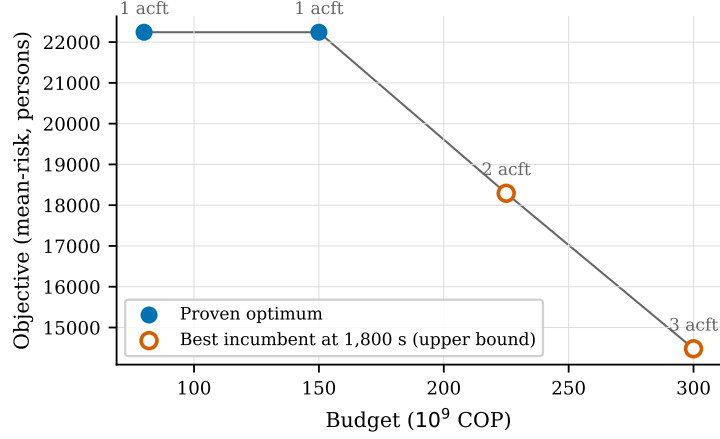


Figure 5: Experiment 6, budget axis: the dominant sensitivity axis. Annotations give the fleet size fielded. Open markers are 1,800 s incumbents (upper bounds on the optimum).

at its limit while its incumbent matched the proven optimum of every other cost point, pure configuration symmetry.

The policy reading is sharper than a flat sensitivity table. At these prices, marginal pesos buy more protection as aircraft with their basing than as any refinement of the ground network; but whether an aircraft buys anything at all is decided by the containment regime, that is, by how large fires are when attack begins, which argues for pairing any acquisition with faster detection-to-dispatch. All of it must be read alongside the disclosed no-intra-day-reuse simplification, which understates single-aircraft capacity and correspondingly overstates the marginal value of additional aircraft.

6.7 The plan on the ground

Solved plans are useful to a planner only as places, so we close the study by reading the base-case optimum operationally (the full solution report is generated by a committed script). The plan opens one base, the Cerritos aerodrome, about 79 km west of Bogotá toward the Magdalena valley, stations the single affordable Firehawk there, and enables water points of which exactly six are ever used: the La Regadera reservoir, Lake Guatavita, the Pantano Redondo and Encatada lagoons, the Romero well (used on two of the ten days), and one waterfall pool, each a named, existing water body from the OpenStreetMap and CAR catalogs. Across the ten day-scenarios the aircraft serves the highest-exposure containable fire of each day, seven containments in all, with delivery margins between 1 and 30 percent of the requirement (30,283 liters delivered against a 30,000 liter requirement on the tightest day): the water-cycle coupling is the binding margin of every single containment, not a theoretical nicety. The worst scenario in the CVaR tail is the real day of 15 January 2024, eighteen simultaneous fires, the very episode that motivated the Firehawk program.

Three operational readings follow, stated with the caveat that several bases tie as alternate optima (Cerritos, Colombaima, and aerodromes as far as Sogamoso and Canaima appear across equivalent runs). First, what is stable across every tie is not the aerodrome but the served fires and the water points that serve them: the six water bodies above cost about 300 M COP in total at the swept enabling cost, 0.4 percent of one aircraft, so securing legal and logistical dip access to a short list of named reservoirs and lagoons is the highest-leverage, lowest-cost action available, and it is actionable before any aircraft arrives. Second, because the base choice is degenerate, the aerodrome can be selected on criteria the model does not price, existing infrastructure, maintenance co-location, military logistics, at no measured loss of protection,

provided the water list comes with it. Third, the containability cliff of Section 6.6 translates into dispatch doctrine: every containment in the plan happens because the aircraft attacks the day’s best containable fire within the two-hour window, and at attack sizes above five hectares nothing is containable, so the value of the acquisition rides on detection-to-dispatch speed as much as on where anything is sited.

7 Conclusions, limitations, and ongoing work

We introduced the first model that jointly sites aerial suppression bases and water refill points with cycle-endogenous aircraft productivity, under a two-stage stochastic program with a mean-risk CVaR objective, gave it an exact linearization and a verified-exact fix-and-optimize matheuristic, and built a fully sourced open-data case study for Cundinamarca around Colombia’s real Firehawk acquisition. The computational study’s contribution is structural: a fleet-first, physics-second regime in which sequential siting is indistinguishable from optimal in every certifiable configuration; containment constants that act as a containability cliff rather than as nuisance parameters, so the model separates the fire regimes an acquisition can and cannot defend; site costs that never bind; a tractability cliff driven by fleet size and budget slack; CVaR degeneracies that argue for the mean-risk form on real data; and an operational reading in real places, a one-aircraft plan whose protection rides on a short, nearly free list of named water bodies and on attack speed rather than on the specific aerodrome chosen (Section 6.7).

Limitations are disclosed throughout and summarized here. The containment law is a phenomenological exponential with uncalibrated, swept constants; scenarios cover a single year’s bootstrap pool, so interannual variability is unrepresented; each scenario is a no-reuse episode, understating per-aircraft daily capacity; the aircraft type is single; site costs are uniform sweep points; and water points are uncapacitated, with no volume depletion or simultaneous-use congestion, a simplification that matters most precisely in the multi-aircraft regimes where the integrated model’s value concentrates. The one open computational question is certification in the three-aircraft, high-budget-slack regime: four independent solution paths agree there to solver tolerance (Section 6.3), but none proves optimality, because budget slack destroys the direct lower bound through configuration symmetry; symmetry breaking or stronger bounds for that regime are the natural next step.

Reproducibility. All code, the data pipeline, every experiment script, and the verification log of every decision and citation are available in the companion repository; raw data are redistributable open sources referenced above.

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